

Quadruped robot: from components to a controlled system

M22 • 03 Sep 2028 - 01 Oct 2028 • 75 hours

This month's deliverable

Assemble and commission a quadruped with 12 degrees of freedom: four legs, each with three actuated joints. The deliverable is **an assembled robot ready for final acceptance**: a specific hardware build, reproducible configuration, calibrations and test results demonstrating at least 60 seconds of standing and a slow gait on a prepared level floor. This stage verifies the architecture before the six-legged version, which remains the programme's final goal.

The software model, mechanical assembly and visualiser must describe the same physical unit. For every actuator, record its physical location, identifier, positive movement direction, zero, operating limits and firmware version. Save raw measurements and conditions for every trial: battery position, mass, surface, IMU mounting, communication frequencies and initial temperature.

What must be ready at entry

Prepare the outputs of earlier modules: a tested leg and actuator rig, CAD and drawings, power diagram, URDF, standing and gait controller, state estimator, diagnostics and failure scenarios. The entry check is to run a saved simulation test and explain the origin of every actuator command. If an axis sign or encoder conversion is unclear, resolve it on one leg first.

List the components actually available and the missing parts. Do not schedule a full-machine trial before the necessary support, wiring and measurement equipment are available. This month's main tools are the existing C++ and ROS 2 software, Python for analysis, a multimeter and suitable current and temperature instruments; choose ranges for your system. There is no need to introduce a new programming platform at this stage.

Scope and connection to the hand

M22 covers assembly, measurement, calibration and transfer of control to real hardware. Running, jumping, difficult terrain and neural-network control are outside the minimum deliverable. Actuator records, current monitoring, hardware motion stopping and calibration logs can be reused for the hand. The hand remains a separate device: successful robot standing does not verify it.

Distinguish three evidence levels: simulation, a supported test rig, and the physical robot on the floor. A suspended test does not establish stable standing. If a hardware problem cannot be resolved this month, save a release candidate with its blockers and continue the repair. Progression to 18 actuators requires the M23 entry check.

Organising 75 hours and the workspace

Four sequential blocks

Hours 1-20: 10 hours studying construction, power and commissioning procedures; 10 hours of assembly, inspection and initial power-up. Deliverables: the unit record and power-verification protocol. **21-40:** 10 hours on calibration and communication; 10 hours measuring one actuator, one leg and then the remaining legs. Deliverables: a machine-readable calibration file and actuator mapping table.

41-60: 10 hours analysing HIL, contact, IMU and delays; 10 hours comparing the model with real measurements and rerunning the controller. **61-75:** 5 hours analysing standing and the test programme; 10 hours practising standing, slow stepping, controlled stopping and preparing the candidate. Total: 35 hours of theory and analysis and 40 of practice; protocol preparation and data processing are included.

Arranging sessions

Organise each 10-hour analysis block into five 2-hour sessions: begin with a specific question, read the relevant section, calculate or inspect a log, then decide what to do on the rig. Split the final 5 hours into 2 + 2 + 1. Each practice block comprises 10 study hours in the Sunday practical, with breaks and alternating assembly, measurement and analysis. Do not operate actuators continuously for 10 hours: use cooling time to process recordings.

The dates on the first page are the calendar window from the original plan. A boundary date may belong to two modules; allocate available sessions without double-counting. Retain rest, holidays and separate integration G blocks from the overall plan. The hour numbering below specifies workload and order; it does not add evening sessions to the calendar.

Rig organisation

Separate the workspace into unpowered assembly, measurement and potential-motion areas. Prepare a stable support that holds the body when actuators are disabled, and provide access to the hardware E-stop. Before starting, specify the permitted motion and stopping condition. Do not use your hands to hold the legs or body as the primary motion restraint.

Each attempt's log contains run_id, author, Git commit identifier, CAD revision, configuration hash, initial conditions, expected behaviour and actual outcome. Record interventions separately: retightening fasteners, replacing a cable or moving the battery. A condition changed during a trial defines a new experiment. Record a missing measurement as no data, not zero current or zero error.

Mechanics, mass and load distribution

What to study

Identify the connections that transmit load from body to foot: frame, hip-actuator mounts, links, shafts, bearings and threaded inserts. Read the insert and fastener manufacturers' guidance for the material used, together with the test results for your printed parts. Check print orientation, local cracks and deformation after tightening. No universal tightening torque applies to every printed connection.

Review centre of mass: $r_{COM} = \Sigma(m_i r_i) / \Sigma m_i$, with masses in kilograms, coordinates in metres and all r_i expressed in one frame. CAD provides an initial estimate, but measure the mass of cables, battery and actual actuators. For quasi-static standing, the centre-of-mass projection and actual contacts help estimate support margin; they do not prove dynamic stability.

Task 1. Record the actual assembly

Inputs are CAD, the parts list, the assembled body and four legs with actuator power disconnected. 1) Assign leg names LF, RF, LR, RR and mark body axes. 2) Weigh assemblies and the complete unit in one configuration. 3) Compare measured mass with the assembly sum and CAD. 4) Inspect every loaded connection, insert and clearance throughout the admissible manual movement range, where the actuator permits manual movement.

Save assembly_manifest.csv, photographs of connections and a discrepancy list. Check that every loaded fastener and insert has an inspection status, and explain the difference between total and summed masses considering scale resolution. Boundary case: a cable is trapped only with the leg bent. Record the collision configuration, reroute the cable and repeat the complete inspection.

Task 2. Check centre of mass and deflection

Use supports or scales with suitable ranges and known contact coordinates. Measure reactions in a repeatable static configuration and estimate horizontal centre-of-mass position from moment balance. Repeat after moving the battery between two intended locations. Do not infer centre-of-mass height from this horizontal-plane measurement; use a model estimate and state its uncertainty.

Save mass_properties.md with calculations, battery positions and a frame-deflection estimate from markers or a measuring instrument. Check that centre-of-mass shift agrees with battery movement and the reaction sum agrees with weight. If a support carries no load, do not count it as a confirmed contact. Reposition supports and repeat the measurement, then update model inertial parameters only from established quantities.

Power, cable harnesses and independent motion stopping

What to review before power-up

Trace the power circuit from the battery or supply through protection, switching and distribution to each actuator. Identify computer and sensor power, common grounds, signal interfaces, cable strain relief and connectors separately. For each branch, list wire specifications, connector ratings, allowable conditions, protective device and its time-current characteristic. Comparing a single current rating is insufficient.

Distinguish controller current limiting, wiring protection, BMS and emergency energy removal. They serve different purposes and do not replace one another. If the circuit uses a contactor, check its suitability for DC, voltage, load characteristics and coil-transient suppression against component documentation. Develop a procedure for the selected circuit here; do not assign generic fuse values.

Task 3. Wiring checks and staged power-up

Inputs are the de-energised assembly, circuit diagram and component instructions. 1) Check continuity of intended connections, absence of unwanted connections between power rails, and connector polarity. 2) Check harness restraint throughout the leg's permitted range. 3) Power the low-power section and measure voltages. 4) Connect power branches using the approved procedure, beginning with one supported actuator; do not enable motion yet.

Save `powerup_checklist.csv` and an oscilloscope trace or log of available voltage and current measurements. Check that measured polarity and ranges match specifications; explain unexpected consumption before the next stage. Boundary case: computer supply voltage sags when an actuator is powered. Return to disconnected actuator power and check inrush current, DC-DC conversion, ground and connections; increasing a software timeout does not repair a power fault.

Task 4. Demonstrate E-stop operation

Define the physically achievable state after an emergency stop: support by an independent structure, removal of actuator energy and prevention of automatic reactivation. On a protected rig, test the circuit with the programme running and with the control process stopped. Record the button-press instant and an independent indication that force or torque application has ceased, using a circuit or actuator measurement appropriate to the design.

Save `estop_test.md` with the circuit, measured response time and residual-motion description. Derive the timing criterion from admissible displacement and mechanism characteristics. Releasing the pressed button alone must not resume an earlier motion command: test recovery separately. If torque removal makes the robot fall, test with support; improve how the safe state is achieved instead of calling free fall a successful stop.

Calibrating 12 actuators and reliable communication

The single-joint model

For each joint, record $q = s \cdot k \cdot (n - n_0)$, where q is angle in radians, n is the encoder count, n_0 is the calibration zero, s is $+1$ or -1 , and k is radians per count including the transmission ratio. The formula applies only to your encoder mode; count wraparound and multiple turns need separate rules. Do not apply another actuator's conversion factor merely because its connector looks similar.

Separate the mechanism's physical limit from the software operating boundary. The latter must account for calibration error, backlash, deformation, stopping distance and cable clearance. A limit recorded in the model still requires verification in the command path. For current and temperature, state measurement source, scale, offset and interpretation: built-in housing or board temperature may differ from winding temperature.

Task 5. Identifiers, signs and zeros

Connect one actuator or one leg with support that constrains unexpected motion. 1) Read ID, model and firmware and match them to the physical label. 2) In the permitted small region, command a known increment and compare motion with the positive URDF axis. 3) Establish a repeatable mechanical reference pose and calculate n_0 . 4) Repeat approaches to the reference from both directions, then apply the procedure to the remaining joints.

Save calibration.yaml and a table of repeats. Check that all 12 IDs are unique, commands reach the observed joint, and residual error and backlash measured in radians remain within a tolerance justified in advance. Boundary cases are a mirrored leg and a swapped cable. The script must detect disagreement with the hardware map; bus response order alone does not identify which actuator to enable.

Task 6. Frequency, errors and bus utilisation

First calculate communication time from byte count, bit rate and framing for the actual interface; add response and direction-switching delays. Then measure the real cycle for 1, 3 and 12 devices with the same command frequency and telemetry content. For DYNAMIXEL, study Sync Read/Write and Status Packet; for another bus, use its primary specification. Do not count a missing reply as packet loss when the protocol specifies no reply to that command.

Save bus_budget.csv, the cycle-duration distribution and counters for timeouts, CRC errors and retries. Calculate margin to the communication deadline using a measured adverse case and list untested modes separately. In the driver simulator, suppress one reply: only that joint's data must become stale, and an old sample must not be republished as new.

To calibrate current, compare the built-in channel with an independent measurement of the same physical quantity in several permitted modes; record scale, offset and error. Check temperature against a reference at a comparable location and in steady conditions. A board sensor cannot be calibrated as a winding sensor by simple subtraction. If the register cannot be corrected, save the verified conversion model and its limitations in the actuator record.

HIL and comparison with real measurements

What HIL means here

HIL is a test in which part of the real computing or hardware chain operates with a simulated plant. Draw the boundary: identify which process, controller, driver, bus and sensor are real and which are replaced. If control hardware is absent, call processing a file on a laptop replay. Match interface versions, units and timestamps; the HIL label alone does not make a test more accurate.

Tabulate discrepancies between model and robot: mass, link lengths, zeros, command delay, filters, friction, backlash, stiffness and foot contact. For each, specify its expected effect and measurement method. Correct geometry and signs first, then timing and dynamics. Simultaneously tuning many coefficients hides the root cause and does not produce reproducible improvement.

Task 7. Time alignment and control verification

Take a low-amplitude target sequence already verified on the leg and one log of its physical response. Record command-generation and transmission times, data-reception time and measurement timestamp if available. Establish the relationship between clocks first; subtracting timestamps from unsynchronised devices does not measure delay. Plot q_{cmd} , q_{meas} and interface events together, estimate the shift, then replay the same input in the model.

Save `fidelity_gaps.csv` and a before-and-after report for one justified parameter correction. Check that input, initial pose and frequency are identical; calculate RMSE on a common time grid and label it in radians. Boundary case: several samples are lost. Mark the gap and do not present interpolated values as complete measurements. Check controller changes by reprocessing the recording dataset before restoring actuator power.

Task 8. IMU and foot contact

With the body secured, check physical axes against `frame_id`, then test known static tilts using an external reference. Verify roll and pitch signs and IMU orientation relative to the body. Under translational acceleration, do not treat tilt calculated from the accelerometer alone as ground truth. For each foot, compare detected contact against observed touching and unloading, using the chosen sensor or an indirect estimate with stated uncertainty.

Save `imu_contact_validation.md`: reference, reference uncertainty, orientation errors and a table of correct and incorrect contact classifications. Control case: a leg touches the floor but slips; contact exists, but there is no stationary support. Remove one contact from estimator input and check the degradation flag. Accompany simulation friction tuning with a surface description; one fitted value does not characterise all real terrain.

Retain the M19 classical algorithm and M20 force allocator; check QP failure and timeout. Before operation, establish whether the actuator supports the required torque or current mode. $J^T f$ is not a ready-to-use position command: a position interface requires a justified inner and outer control-loop design. Keep the M21 policy disabled during initial integration.

Integrated laboratory: standing and the first step

Entry conditions and protocol

The assembly may enter this laboratory after tasks 1-8: known calibrations, verified independent stopping, working diagnostics, and a recording and time-alignment procedure consistent with measurements. Prepare a level area, foot and body markers, a backup support that carries no weight during the qualifying stand, a way to observe displacement, and all measurable limits. Record initial temperature, charge, voltage and battery configuration.

Before starting, specify operating boundaries: body-position region, joint limits, allowable data age, current, temperature and trial-termination conditions. For every threshold, provide its calculation or specification, unit and response. Criteria must use measurable quantities rather than descriptions such as not very hot or a plot colour. Removing torque and controlled lowering are different actions, selected according to the stopping cause and mechanism capability.

Task 9. Stand for at least 60 seconds

1) On the support, verify coordinated movement of all four legs to the permitted starting pose. 2) Place feet on their marked locations and gradually transfer load to the robot using your procedure. 3) Run a short trial and inspect currents and contact. 4) Only then run a continuous qualifying interval of at least 60 seconds. Record roll and pitch, body height and displacement where measurable, joint errors, currents, temperatures and events.

Save `stand_report.md`, video showing the backup support and the raw log. Success means independent standing throughout the interval without crossing declared boundaries or hiding telemetry losses. Calculate mean and maximum absolute roll and pitch errors separately and state units explicitly. If the backup support carries weight or an operator supports the body, the attempt does not establish standing. Do not remove that moment from the evidence.

Task 10. Slow gait and stopping

After standing, perform a limited crawl sequence: transfer the load centre, lift one foot, move it and confirm touchdown. Begin with one cycle, then attempt a short marked route. Compare initial and final poses, completed steps and contact losses. A commanded stop must reach the previously described state; an emergency cause uses the independent stopping path.

Save `crawl_run` and `stop_run` with video aligned to logs. Repeat a small, bounded change in tilt or load under identical conditions, first on the rig; derive method and amplitude from allowable loads instead of applying a hand strike. Check for recovery into the declared region within a measured time, or a timely stop. One successful traversal is a demonstration, not proof of reliability.

Separately from the qualifying stand, command a small permitted sequence of body roll and pitch changes, save target and measurement, and calculate error and settling time. Set amplitude from available kinematics and loading. This tests command tracking as well as holding one pose; stop the experiment on joint saturation according to the adopted criterion.

Failure tests and data validity

Six required tests

T1 - incorrect ID or sign. Substitute a configuration in the test environment, then verify it by reading hardware on the rig with actuator power disconnected. Activation must be refused with the exact joint name. **T2 - stale sample.** Repeat the last value with its original timestamp in the driver simulator. The controller must distinguish a missing new measurement from a stationary joint.

T3 - missing reply. Simulate one lost packet, then a sequence of losses; check counters and state transitions without an automatic hidden reset. **T4 - command beyond the limit.** Supply an inadmissible target only to the software path or HIL; verify rejection or documented clamping before writing to the actuator. This test does not require contact with the mechanical stop.

T5 - emergency stop with a stalled programme. With the robot on a support, stop the control process and check the independent path for ending force or torque application. **T6 - inconsistent contact.** During replay, supply contact during swing or loss of expected support; the system must report degradation and enter the intended state. Save the input, expected event and observed sequence for every test.

Heat, current and physical repeats

Also test slowly rising temperature and current indication beyond the set threshold by substituting telemetry in HIL. This establishes response logic; a separate circuit-specific rig test establishes the protection's physical ability to remove energy. Do not create a short circuit or overheat a motor to check a software flag. Load tests remain within the previously calculated range.

State which current is measured: supply current, branch current, phase current or the actuator's internal estimate. Quantities with different definitions cannot be summed without a model. For temperature, identify sensor location and sampling interval. Failure to record a peak with infrequent sampling does not prove there was no peak. Telemetry losses make the relevant metric incomplete.

Documenting verification

One test_matrix.csv row contains test ID, revision, evidence level, preconditions, stimulus, expected result, measured result, PASS/FAIL/INCOMPLETE and log link. Run required scenarios first in the simulator, then only those necessary and permissible on the rig. Do not reproduce physical faults by dangerously destroying components.

Do not exclude a failed attempt from statistics without a reason defined before the trial. A startup failure caused by your programme is a result, not bad data. An external-instrument error may invalidate a trial, but retain the record. Fully analyse at least one real failure or deviation and its correction: symptom, hypothesis, check, change and repeat.

The first 40 hours: detailed sequence

Hours 1-10: five 2-hour analysis sessions

1-2: reread the M22 objective, run the original algorithm in simulation or replay and inventory earlier-module outputs. 3-4: trace load transmission in CAD and list questionable connections and task 1 measurements. 5-6: calculate centre of mass and plan task 2 reaction measurements; state method limitations. 7-8: annotate the power diagram, ratings and unknown data, and prepare continuity testing. 9-10: describe the E-stop, support and restart prevention; rehearse without motion.

Use the final 15 minutes of each session to record decisions. If a component rating is unknown, name who will clarify it and the document required. An assumption must not become permission to power up. Before the practical, prepare a short action sheet with power-up and stopping sequences, available beside the rig.

Hours 11-20: the first practical

11-13: assembly, labels, inspection and weighing for task 1. 14-15: reaction measurements and centre-of-mass estimation for task 2. 16-18: wiring, continuity checks and staged power for task 3. 19-20: independent-stop verification for task 4 and evidence recording. These intervals include instrument setup and documenting results; repeats after finding a defect use time from the same block.

If the E-stop fails verification, do not start the next motion stage. Use remaining time to repair the circuit and make passive measurements. The month may end with a candidate that has not yet passed; retain that status instead of recording an unsupported pass. Calendar pressure does not relax the mandatory deliverable.

Hours 21-40: calibration and the bus

21-22: encoder conversion and zero definition. 23-24: mirrored legs, axis signs, operating limits and uncertainty. 25-26: physical meaning of current and temperature readings for the selected actuator. 27-28: calculate frame size and bus utilisation. 29-30: hardware-component lifecycle, read-error handling and pre-activation checks. These are five 2-hour analysis sessions.

31-34: calibrate one leg and then the others; record zero-repeatability results. 35-37: compare communication with 1, 3 and 12 devices and check timeouts in the simulator. 38-39: parameterised hardware-map and limit tests. 40: inspect after first activation, freeze the calibration version and prepare a log for model comparison. By the end of this block, physical names and software indices must map unambiguously.

Hours 41-75: from HIL to a release candidate

Hours 41-50: transfer analysis

41-42: define the HIL boundary and tabulate timing in command and measurement paths. 43-44: estimate delay and comparison limitations from task 7. 45-46: check measured masses, lengths and contact parameters; choose one significant model correction. 47-48: review IMU conventions, the tilt reference and contact-trust rules. 49-50: plan estimator and controller regression checks using original and corrupted logs.

At this stage, read only the required primary-source sections on page 12. Reading must produce a revised diagram, test or calculation. If dynamics material extends substantially beyond your control stack, document the model limitation and retain the previously verified controller; this month integrates material already studied.

Hours 51-60: the third practical

51-53: obtain a low-amplitude physical response from one leg and align it in time. 54-55: correct one parameter, repeat the same input and record improvement or deterioration. 56-57: measure the IMU in reference orientations and verify foot contact. 58-59: run T1-T6 in software or HIL and verify recovery. 60: prepare the complete standing configuration and reinspect harnesses after movement.

Do not spend the entire block seeking visually perfect simulation-hardware agreement. Establish the magnitude of significant discrepancies and verify that the controller respects limits in their presence. Record at least two explained discrepancies and one still unexplained, if present. An incomplete model is a limitation that can be investigated experimentally.

Hours 61-75: completion

61-62: define standing and roll/pitch tracking criteria, stopping conditions and measurement methods. 63-64: specify the slow gait, controlled stop and repeatable small disturbance. 65: review readiness from documentation; state which result permits or blocks M23. These 5 hours are split into 2 + 2 + 1.

66-68: preliminary and qualifying standing for task 9, with analysis between attempts. 69-71: a slow gait cycle, short route and normal stop for task 10. 72-73: permissible disturbance repeats and checks for new defects. 74-75: assemble the candidate package, verify plot reproduction and complete the progression decision. If time is short, omit video editing, additional routes and optional optimisation; calibrations, E-stop, standing and raw trial records remain mandatory.

M22 acceptance and handoff to the six-legged robot

What complete means

The release candidate contains source code and the exact assembly revision; a circuit with ratings and selection rationale; mass and centre of mass with measurement descriptions; the 12-actuator map; calibrations and operating limits; hardware-stop documentation; a consistent URDF; HIL, replay, IMU and contact records; and evidence of independent standing for at least 60 seconds, slow walking and controlled stopping. During the completed trials, currents, temperatures and communication parameters remain within operating ranges specified in advance.

Turn the original criterion, no unbounded growth of current, temperature or bus errors, into an observable check: every signal has a source, sampling interval, limit, response and test duration. A short normal run does not authorise continuous operation of unknown duration. An unresolved cause of control loss or an unreliable emergency stop blocks physical scaling.

Engineering self-check

Answer without running the programme: 1) Which joint moves when a chosen ID is addressed, and why? 2) How was its zero measured, and what is the spread? 3) What happens after emergency torque removal? 4) Which part of stopping is independent of the main computer? 5) How does a measurement timestamp differ from recording time? 6) Why does supply current not equal torque at each joint? 7) How do a new measurement from a stationary sensor and a stale sample differ? 8) Which simulation-reality discrepancy constrains the next trial?

Practical defence: choose one failed run, open its raw log, reproduce the plot and identify the root cause or explicitly state the hypothesis still untested. Show which change tests it. Then explain the power-disconnection chain on the diagram and the axis direction using a physical label. A screenshot of a successful simulation cannot replace this check.

The M23 package and the separate hand workstream

Create handoff_M23.md: verified quadruped capabilities, test boundaries, problematic parts, maximum observed bus utilisation and a projection for 18 actuators. Attach the progression decision PASS/HOLD. PASS means the scaling entry requirements are met; it does not establish that battery, frame and bus are already suitable for six legs.

For the hand, pass on the calibration.yaml template, diagnostic scheme and method of linking evidence to a physical unit. Do not copy numerical leg limits to fingers. The hand and tactile-equipment work in OPT 120 hours remains a separate workstream begun after M17. M22 does not consume its time or present its results as results of this assembly. Completion of both systems is checked at the end of the overall programme.

Assigned reading: quadruped assembly and measurement

The references may be in English; the exercises and definitions are presented here in English. Read only the named sections before the associated work, within the existing 75 hours.

Core reading

1. K. Lynch, F. Park - Modern Robotics. Sections 4.1.1 and 8.9: forward kinematics in the space frame; actuation, gearing and friction. For the 12-joint map, connect physical zero and sign to the model; for current tests, explain the effects of transmission and friction.

2. NASA - Systems Engineering Handbook. 5.2 Product Integration; 5.3 Product Verification; 5.4 Product Validation. Before HIL and floor trials, separate evidence of assembly, compliance with requirements and suitability of the selected mode. Deliverable: the power-up sequence and entry criteria for each rig.

Reference book: power problems

3. P. Horowitz, W. Hill - The Art of Electronics, 3rd ed. 9.4 Heat and power design; 9.6 Switching regulators and dc-dc converters. Read selected heating and power-conversion calculations for task 4. The aim is to understand losses and limits of a ready-made DC-DC converter; designing a mains supply is unnecessary.

Documents before specific checks

ros2_control, Jazzy. Hardware Components: interface types, lifecycle, errors. For the driver, specify which states and errors permit command output; relate this to independent stopping.

ROS REP-103 / REP-145. Units, Axis Orientation, Rotation Representation; IMU: Data Reporting, Common Parameters. For axis and IMU calibration, state the coordinate frame, units and meaning of acceleration; perform static verification before standing.

The actual bus specification. For DYNAMIXEL Protocol 2.0: Status Packet, Sync Read/Write, Bulk Read. For another actuator, use its specification. Calculate communication volume with 12 devices and the cycle deadline; a similar connector does not establish compatibility.

Separately save specification pages for your actuators, battery, BMS, charger, DC-DC converter and protective devices: voltage, current, temperature, mode and duration. A textbook example does not establish this assembly's operating limits.

Terms and definitions

Commissioning. Connecting and verifying an assembled device in stages, progressing to the next mode only after declared criteria are met.

Hardware revision. An identifier for the exact composition and implementation of an assembly: parts, wiring, sensors and board versions to which measurements apply.

Actuator map. A table linking physical joints to bus addresses, directions, zeros, units and limits; verified by real movement of every assembly.

Zero calibration. Determining the offset between a sensor reading and a selected geometric zero, recording the method, conditions and repeatability spread.

Backlash. Relative movement within a transmission when load direction reverses, during which input motion is not transmitted to the output.

Continuous torque. Torque admissible at specified speed, cooling and sustained thermal conditions; it must not be equated with short-duration stalled-shaft torque.

Centre of mass. The mass-weighted position of the complete assembly; it changes as legs move or the battery is repositioned.

Inertia tensor. A matrix describing mass distribution about a selected point and axes; it relates angular motion to torques, with units $\text{kg}\cdot\text{m}^2$.

Duty cycle. The fraction of time under active load within a declared cycle; together with peaks and cooling, it affects actuator heating.

Power tree. A diagram of sources, converters, protective devices and loads showing current paths and points where individual branches can be disconnected.

Voltage sag. A drop in voltage under load caused by source and wiring resistance or supply limiting; it can reset electronics.

BMS. A system monitoring battery cells and providing specified protection; it does not replace a compatible charger, fuses and emergency stop.

DC-DC. A DC voltage converter selected for input range, output voltage and current, losses, cooling and response to transient loads.

E-stop. A specified independent means of ending hazardous motion; the structure may fall after torque removal, so verify the final state.

Watchdog. Monitoring timely confirmation of correct operation; exceeding the timeout triggers a predefined response independently of the normal command cycle.

Communication deadline. The latest time for obtaining required data and issuing a command; exceeding it requires a response even without a packet error.

HIL. Testing in which real hardware interacts with a simulated part of the system; explicitly document which channels are real.

Specific force. The ideal accelerometer reading associated with a $-g$ in the sensor frame; it is not directly equal to body acceleration in world coordinates.

Relationships: $P = UI$; conductor losses $P = I^2R$. Check: disconnecting actuator power and reaching a safe final state are different facts.

Quadruped purchases and setup

USE EXISTING EQUIPMENT

Use the tested 3-degree-of-freedom leg, its bus interface, MCU and IMU, RGB-D camera, multimeter, CC/CV supply, analyser, tools and Mac M4. For the existing 3D printer, select a profile for its exact model; replenish consumables and fasteners according to released CAD and BOM.

BUY BEFORE ASSEMBLY - AFTER LEG ACCEPTANCE

9 identical working actuators plus **2 separate spares**: $3 + 9 = 12$ installed. The family, voltage, bus, thermal regime and continuous loading have already been verified on the leg. Add parts for three legs, frame, harnesses and fasteners according to the BOM; the M7 bench DC motor is not included in these 12.

1 onboard computer: Pi 5 with 8 GB or an Ubuntu-compatible x86 computer after measuring CPU, RAM and USB loads. Reuse a suitable computer already purchased. The Pi needs 1 boot medium, cooling and power according to [Pi documentation](#); account for camera consumption separately. The model name does not guarantee every driver will work.

1 power system: battery, compatible charger and BMS, hardware stop and power disconnection, branch fuses, DC-DC converter, connectors and wiring. Select ratings after measuring operating and peak currents, duration, voltage sag and heating; the bench supply limit is not the power budget for 12 actuators. The number of protective devices matches the branch count.

1 support or backup frame with verified load capacity; **1 IR thermometer** with a suitable range and allowance for surface properties; spare harness components. If choosing contact-sensing feet, provide **4 channels, one per foot**, repeatable activation and calibration for robot mass; these sensors do not establish absence of slip.

INSTALL AND VERIFY

Onboard, use [Ubuntu 24.04 ARM64 for Pi](#) or [24.04 x86_64](#); [ROS 2 jazzy](#), existing `ros2_control`, bus driver and diagnostics from the pinned working environment. Retain the existing environment on the Mac and in UTM.

Check the build, software simulator and replay without motion; 12 unique IDs; units and sign of one supported channel; heartbeat loss; current and temperature recording; and independent stopping. Test USB, network and camera under load on the chosen computer. Do not enable torque automatically when the OS starts.

For the model: [URDF inertial/collision](#); for logs: [Foxglove Plot](#); for the ID table: [pytest parametrization](#). Preparation and checks fit within the existing 75 hours, replacing repeated reading and setup. Delivery waiting time does not count as study time.

Motion fundamentals: loading before the first step

Required analysis: measured reactions, centre of mass and model limits

Kinematics shows where a foot can be placed. Forces and torques are needed to determine whether the robot can withstand standing and acceleration. Begin with measurable statics: it reveals errors in mass, centre-of-mass and load-transmission estimates before gait tuning. The number of supports alone does not imply equal weight distribution.

Free-body diagram and equilibrium

The example body model assumes a horizontal surface, a stationary robot, gravity and vertical reactions from two support groups. The x axis points forwards; the rear group is at $x=-a$, the front at $x=+a$; z points upwards. F_r and F_f are ground forces on the robot in N, and x_c is centre-of-mass position. Force and moment balance give $F_r+F_f=mg$; $a(F_f-F_r)=mg \cdot x_c$. Therefore, $F_f=mg(a+x_c)/(2a)$, $F_r=mg-F_f$.

Example. $m=4$ kg, $a=0.20$ m, $x_c=0.05$ m, $g=9.81$ m/s²: $F_f=24.525$ N, $F_r=14.715$ N. Adding a 0.5 kg load at $x=0.25$ m gives total mass 4.5 kg and $x_c=(4 \cdot 0.05 + 0.5 \cdot 0.25)/4.5=0.07222$ m. The new reactions are 30.043 N and 14.102 N. Rear loading decreases: the payload adds a moment as well as weight.

Practice: verify the calculation on the assembled robot

1. Measure mass and grouped reactions in task 2 using existing fixtures and instruments. If an instrument displays equivalent mass in kg, convert it to N using g. Recover $x_c=a(F_f-F_r)/(F_f+F_r)$, compare against CAD and include the mass of cables, battery and mounts. Two sums cannot determine the force at each of four feet.

2. Relate load to the joint. For one adopted pose from M8, calculate J and the force in the same coordinate frame; the force contribution is $\tau_{\text{ext}}=J^T F_{\text{ext}}$. During holding, the actuator must compensate for it while accounting for link weight; compare with the M9 model. Convert current to torque only through a calibrated transmission model. Do not send a calculated torque to a position-controlled servo as an angle.

What counts as evidence

Save load_balance.csv: pose, payload, reactions, calculation, residuals, units and instrument uncertainties. Check reaction sum, moment balance and direction of centre-of-mass displacement. Set tolerance from scale resolution, repeatability and geometry; do not impose 1% without justification. A negative calculated reaction means that support is infeasible, not that the foot is attracted to the floor.

Within 75 hours: 1 hour of analysis in 5-6 and the practical in 14-15 (task 2), without a new motion mode. This is a quasi-static check: accelerations, slopes and external disturbances require the full model. Read [Modern Robotics §5.2](#) - the relationship between external force and joint torques.